

Feiyang Wang

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DoB : May 2003 | MRes AI & ML, Imperial College London

EDUCATION BACKGROUND

Imperial College London

London, UK

MRes in Artificial Intelligence and Machine Learning

Sept 2025 – Sept 2026

(Expected)

Supervisor: Prof. Chen Qin, Imperial College London

Research: Trustworthy AI with uncertainty modeling; dynamic MRI reconstruction with flow matching; uncertainty propagation through data consistency (DC) layers.

Central South University

Changsha , CHINA

Bachelor of Artificial Intelligence

Sept 2021 - Jul 2025

Average Score: 86.5

Key Courses : Biometric Recognition, Natural Language Processing, Robotics, Medical Image Processing, Image Engineering, Artificial Intelligence, Deep Learning, Pattern Recognition and Machine Learning.

PUBLICATIONS

- **Wang F**, Luo N, Wu W. VisionCube: 3D-Aware Vision-Language Model for Multi-Step Spatial Reasoning[C]//Proceedings of the Computer Vision and Pattern Recognition Conference. 2025: 3270-3279.
- **Wang F**, Yu X, Wu W. CubeRobot: Grounding Language in Rubik's Cube Manipulation via Vision-Language Model[C]//Companion Proceedings of the ACM on Web Conference 2025. 2025: 2181-2186.
- Yu X, **Wang F**, Qiao Z. SpikEmo: Enhancing Emotion Recognition With Spiking Temporal Dynamics in Conversations[C]//Proceedings of the International Joint Conference on Neural Networks. 2025: 1-7.
- Luo X*, **Wang F***, Qin C. Unsupervised Dynamic Parallel MRI Reconstruction via Length-Extrapolative Spatiotemporal Flow Matching [Early Accept at MICCAI 2026].

INTERNSHIP EXPERIENCE

Shanghai Lianheng Technology Co., Ltd.

Research Intern

July 2023 – August 2023

- Supported large-model R&D by running evaluation experiments, comparing model performance and efficiency, and preparing technical documentation and research reports.

Shanghai Zhenghe Qisheng Corporation

Research Intern

Jul. 2024 – Aug. 2024

- Assisted large-model fine-tuning research by collecting and annotating training data, running algorithm experiments, and summarizing results for model-performance improvement.

Kunshan Japp Software Technology Co., Ltd.

Research Intern

Aug. 2024 - Sept. 2024

- Designed and implemented an intelligent line-following car prototype, integrating sensor-based path detection, embedded control logic, motor/PWM control, and iterative track testing to improve autonomous navigation stability.

PROJECT EXPERIENCES

Unsupervised Dynamic Parallel MRI Reconstruction

Sept2025 – Nov2026

- Proposed an unsupervised framework for accelerated cardiac MRI reconstruction via Projected Conditional Flow Matching (PCFM), requiring no ground-truth fully sampled data; training is conducted entirely on undersampled multi-coil k-space measurements from the CMRxRecon 2023 & 2024 cine datasets (LAX/SAX views, 221 training volunteers).
- Designed the Spatiotemporal Flow U-Net (ST-Flow U-Net) with factorized (2+1)D convolutions for memory-efficient spatiotemporal modeling, periodic Gaussian kernel temporal embeddings for cardiac phase encoding, and ALiBi-augmented factorized spatiotemporal attention for length-extrapolative inference beyond the training sequence length.
- Demonstrated robust reconstruction quality at high acceleration factors with strong out-of-distribution generalization across multi-center, multi-scanner datasets from the CMRxRecon 2025 challenge. Work received early acceptance at MICCAI 2026.

MRI Reconstruction with Uncertainty Quantification via Flow Matching

Sept 2025 – Present

- Developing a supervised flow matching framework for MRI reconstruction in which the model predicts both the image estimate and an uncertainty parameter (sigma); analytically deriving how sigma propagates through each ODE integration step and data consistency (DC) projection layer across unrolled reconstruction iterations.

- Deriving closed-form uncertainty propagation formulas through the DC layer orthogonal projection operator, yielding a spatially resolved uncertainty map at the final reconstruction; aimed at providing clinically interpretable confidence estimates for accelerated MRI under varying acquisition conditions.

SDXL Manga Generator (Generative AI Course Project)

Sept 2025-Nov2025

- Developed an end-to-end pipeline for generating audio-narrated multi-panel manga from short text prompts; utilized Qwen-3-8B for automated story expansion and CLIP-optimized prompt engineering, driving a custom Ghibli-style LoRA-fine-tuned SDXL model for stylized panel synthesis.
- Integrated IP-Adapter FaceID for cross-panel character identity preservation, LBPCASCADE anime face detection with SDXL inpainting for face restoration, and Edge TTS for audio narration; the modular architecture produces cohesive narrative, visual, and auditory output from a single short prompt.

Vision-Language Models for Rubik's Cube Manipulation

Sept 2023 – Aug 2025

- Phase 1 (CubeRobot, WWW 2025): Introduced CubeRobot, a VLM for 3x3 Rubik's Cube solving featuring a dual-loop Chain-of-Thought (CoT) architecture with an outer-loop for hierarchical task decomposition and an inner-loop for iterative execution and self-reflection; integrated a Memory Stream for history-guided replanning. Constructed CubeCoT, a benchmark of 43 annotated subtasks across three difficulty levels, providing a structured evaluation framework for embodied reasoning tasks.
- Phase 2 (VisionCube, CVPRW 2025): Extended the system with Cube3D, a 3D representation learning pipeline combining Instant-NGP for multi-view neural reconstruction, PointNet for point cloud feature extraction with sinusoidal positional encoding, Perceiver for fixed-length latent encoding, and Point Transformer for structured 3D feature learning; introduced an Embodied-Projector fusing 3D features with language tokens via cross-modal attention layers. Significantly outperformed GPT-4V, Qwen2-VL, and CubeRobot baselines; deployed in the RoboGuide virtual environment with a simulated LR Mate 200iD robotic arm for closed-loop action generation and verification.

SKILLS & AWARDS

Computer Skills: C, C++, MATLAB, LaTeX, SQL, Python, PyTorch; computer vision, 3D perception, vision-language/embodied AI, task planning, robot simulation, and embedded control.

Language Skills: English (TOEFL: 95), Mandarin (native), Japanese (N1 level)

Awards:

- Awarded the First-class Outstanding Student Scholarship by Central South University for the 2021–2022 academic year, in recognition of academic excellence and ranking within the top 5% of the cohort.
- Awarded the National Encouragement Scholarship by Central South University for the 2022–2023 academic year.
- Received the Third-class Scholarship in 2023.
- Awarded Third Prize in the National College Student English Competition in 2024.